12 General Renormalization Theory

We saw in the previous chapter that calculations in quantum electrodynamics involving one-loop graphs yield divergent integrals over momentum space, but that these infinities cancel when we express all parameters of the theory in terms of "renormalized" quantities, such as the masses and charges that are actually measured. In 1949 Dyson [1] sketched a proof that this cancellation would take place to all orders in quantum electrodynamics. It was immediately apparent (and will be shown here in Sections 12.1 and 12.2) that Dyson's arguments apply to a larger class of theories with finite numbers of relatively simple interactions, the so-called renormalizable theories, of which quantum electrodynamics is just one simple example.

For some years it was widely thought that any sensible physical theory would have to take the form of a renormalizable quantum field theory. The requirement of renormalizability played a crucial role in the development of the modern "standard model" of weak, electromagnetic, and strong interactions. But as we shall see here, the cancellation of ultraviolet divergences does not really depend on renormalizability; as long as we include every one of the infinite number of interactions allowed by symmetries, the so-called non-renormalizable theories are actually just as renormalizable as renormalizable theories.

It is generally believed today that the realistic theories that we use to describe physics at accessible energies are what are known as "effective field theories." As discussed in Section 12.3, these are low-energy approximations to a more fundamental theory that may not be a field theory at all. Any effective field theory necessarily includes an infinite number of non-renormalizable interactions. Nevertheless, as discussed in Sections 12.3 and 12.4, we expect that at sufficiently low energy all the non-renormalizable interactions in such effective field theories are highly suppressed. Renormalizable theories like quantum electrodynamics and the standard model thus retain their special status in physics, though for reasons that are somewhat different from those that originally motivated the assumption of renormalizability in these theories.

12.1 Degrees of Divergence

Let us consider a very general sort of theory, containing interactions of varying types labelled i. Each interaction may be characterized by the number n_{if} of fields of each type f, and by the number d_i of derivatives acting on these fields.

We will start by calculating the "superficial degree of divergence" D of an arbitrary connected one-particle irreducible Feynman diagram in such a theory. This is the number of factors of momentum in the numerator minus the number in the denominator of the integrand, plus four for every independent four-momentum over which we integrate. The superficial divergence is the actual degree of divergence of the integration over the region of momentum space in which the momenta of all internal lines go to infinity together. That is, if $D > 0$, then the part of the amplitude where all internal momenta go to infinity with a common factor $\kappa \to \infty$ will diverge like

$$\int^{\infty} \kappa^{D-1}\, d\kappa. \tag{12.1.1}$$

In the same sense, an integral with degree of divergence $D = 0$ is logarithmically divergent, and an integral with $D < 0$ is convergent, at least as far as this region of momentum space is concerned. We will come back later to the problem posed by subintegrations that behave worse than the integral over this region.

To calculate D, we will need to know the following about the diagram:

$$\begin{aligned} I_f &\equiv \text{number of internal lines of field type } f, \\ E_f &\equiv \text{number of external lines of field type } f, \\ N_i &\equiv \text{number of vertices of interaction type } i. \end{aligned}$$

We will write the asymptotic behavior of the propagator $\Delta_f(k)$ of a field of type f in the form

$$\Delta_f(k) \sim k^{-2+2s_f}. \tag{12.1.2}$$

Looking back at Chapter 6, we see that $s_f = 0$ for scalar fields, $s_f = \frac{1}{2}$ for Dirac fields, and $s_f = 1$ for massive vector fields. More generally, it can be shown that for massive fields of Lorentz transformation type (A, B), we have $s_f = A + B$. Speaking loosely, we may call s_f the "spin." However, dropping terms that because of gauge invariance have no effect, the effective photon propagator $\eta_{\mu\nu}/k^2$ has $s_f = 0$. A similar result applies to a massive vector field coupled to a conserved current, provided the current does not depend on the vector field. It can also be shown that, in the same sense, the graviton field $g_{\mu\nu}$ has a propagator also with $s_f = 0$.

According to Eq. (12.1.2), the propagators make a total contribution to D equal to

$$\sum_f I_f(2s_f - 2). \tag{12.1.3}$$

Also, the derivatives in each interaction of type i introduce d_i momentum factors into the integrand, yielding a total contribution to D equal to

$$\sum_i N_i d_i. \tag{12.1.4}$$

Finally, we need the total number of independent momentum variables of integration. Each internal line can be labelled with a four-momentum, but these are not all independent; the delta function associated with each vertex imposes a linear relation among these internal momenta, except that one delta function only serves to enforce conservation of the external momenta. Thus, the momentum space integration volume elements contribute to D a term

$$4\left[\sum_f I_f - \left(\sum_i N_i - 1\right)\right], \tag{12.1.5}$$

which, of course, is just four times the number of independent loops in the diagram. Adding the contributions (12.1.3), (12.1.4), and (12.1.5), we find

$$D = \sum_f I_f(2s_f + 2) + \sum_i N_i(d_i - 4) + 4. \tag{12.1.6}$$

Eq. (12.1.6) is not very convenient as it stands, because it gives a value for D that seems to depend on the internal details of the Feynman diagram. Fortunately, it can be simplified by using the topological identities

$$2I_f + E_f = \sum_i N_i n_{if}. \tag{12.1.7}$$

(Each internal line contributes two of the lines attached to vertices, while each external line contributes only one.) Using Eq. (12.1.7) to eliminate I_f, we see that Eq. (12.1.6) becomes

$$D = 4 - \sum_f E_f(s_f + 1) - \sum_i N_i \Delta_i, \tag{12.1.8}$$

where Δ_i is a parameter characterizing interactions of type i:

$$\Delta_i \equiv 4 - d_i - \sum_f n_{if}(s_f + 1). \tag{12.1.9}$$

This result could have been obtained by simple dimensional analysis, without considering the structure of Feynman diagrams. The propagator of a field is a four-dimensional Fourier transform of the vacuum expectation value of a time-ordered product of a pair of free fields, so a conventionally normalized field f whose dimensionality[1] in powers of momentum is $\mathcal{D}_f$ will have a propagator of dimensionality $-4 + 2\mathcal{D}_f$. Hence if the propagator behaves like k^{-2+2s_f} when k is much larger than the mass, then the field must have a dimensionality with $-4 + 2\mathcal{D}_f = -2 + 2s_f$, or $\mathcal{D}_f = 1 + s_f$. An interaction i with n_{if} such fields and d_i derivatives will then have dimensionality $d_i + \sum_f n_{if}(1 + s_f)$. But the action must be dimensionless, so each term in the Lagrangian density must have dimensionality $+4$ to cancel the dimensionality -4 of d^4x. Hence the interaction must have a coupling constant of dimensionality $4 - d_i - \sum_f n_{if}(1 + s_f)$, which is just the parameter Δ_i. The momentum space amplitude corresponding to a connected Feynman graph with E_f external lines of type f is the Fourier transform over $4\sum_f E_f$ coordinates of a vacuum expectation value of the time-ordered product of fields with a total dimensionality $\sum_f E_f(1 + s_f)$, so it has dimensionality $\sum_f E_f(-3 + s_f)$. Of this dimensionality, -4 comes from a momentum space delta function, and $\sum_f E_f(-2 + 2s_f)$ is the dimensionality of the propagators for the external lines, so the momentum space integral itself together with all coupling constant factors has dimensionality

$$\sum_f E_f(-3 + s_f) - (-4) - \sum_f E_f(-2 + 2s_f) = 4 - \sum_f E_f(s_f + 1).$$

The coupling constants for a given Feynman graph have total dimensionality $\sum_i N_i \Delta_i$, leaving the momentum space integral with dimensionality $4 - \sum_f E_f(s_f + 1) - \sum_i N_i \Delta_i$. As long as we are interested in the region of integration where all momenta go to infinity

[1] In this chapter, "dimensionality" will always refer to the dimensionality in powers of mass or momentum, in units with $\hbar = c = 1$. We are using fields that are conventionally normalized, in the sense that the term in the free-field Lagrangian with the largest number of derivatives (which determines the asymptotic behavior of the propagator) has a dimensionless coefficient.

Table 12.1. Terms in the Lagrangian density for quantum electrodynamics. Here d_i, $n_{i\gamma}$, and n_{ie} are the numbers of derivatives, photon fields, and electron fields in the interaction, and Δ_i is the dimensionality of the corresponding coefficient. (Recall that $s_\gamma = 0$, $s_e = \frac{1}{2}$.)

Interaction	d_i	$n_{i\gamma}$	n_{ie}	Δ_i
$-ie\bar{\psi}\not{A}\psi$	0	1	2	$4-1-3=0$
$-\frac{1}{4}(Z_3-1)F_{\mu\nu}F^{\mu\nu}$	2	2	0	$4-2-2=0$
$-(Z_2-1)\bar{\psi}\not{\partial}\psi$	1	0	2	$4-1-3=0$
$[-(Z_2-1)m + Z_2\delta m]\bar{\psi}\psi$	0	0	2	$4-3=1$

together, the degree of divergence of the momentum space integral is its dimensionality, thus justifying Eq. (12.1.8).

If all interactions have $\Delta_i \geq 0$, then Eq. (12.1.8) provides an upper bound on D that depends only on the numbers of external lines of each type, i.e., on the physical process whose amplitude is being calculated:

$$D \leq 4 - \sum_f E_f(s_f + 1). \tag{12.1.10}$$

For example, in the simple version of quantum electrodynamics studied in the previous chapter, the Lagrangian included terms of the types shown in Table 12.1. All interactions here have $\Delta_i \geq 0$, and hence a Feynman diagram with E_γ external photon lines and E_e external Dirac lines will

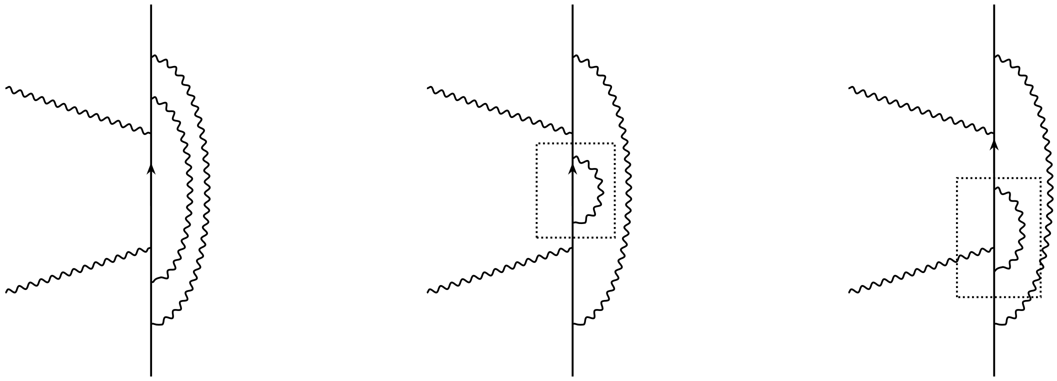
have superficial degree of divergence bounded by Eq. (12.1.10):

$$D \leq 4 - \frac{3}{2}E_e - E_\gamma. \tag{12.1.11}$$

Only a finite number of sets of external lines can yield superficially divergent integrals; these will be enumerated in Section 12.2. We are going to show that the limited number of divergences that appear in theories with $\Delta_i \geq 0$ for all interactions are automatically removed by a redefinition of a finite number of physical constants and a renormalization of fields. For this reason, such theories are called renormalizable. In Section 12.3 we will catalog all the renormalizable theories, and discuss the significance of renormalizability as a criterion for physical theories.

The term "renormalizable" is also applied to individual interactions. Renormalizable interactions are those with $\Delta_i \geq 0$, whose coupling constants have positive or zero dimensionality. Sometimes one distinguishes between interactions with $\Delta_i = 0$, called simply renormalizable, and those with $\Delta_i > 0$, called superrenormalizable. Since adding additional fields or derivatives always lowers Δ_i, there can only be a finite number of renormalizable interactions involving fields of any given types. We have seen that all the interactions in the simplest version of quantum electrodynamics are renormalizable, with the $\bar{\psi}\psi$ terms superrenormalizable.

On the other hand, if any interaction has $\Delta_i < 0$, the degree of divergence (12.1.8) becomes larger and larger the more such vertices we include. No matter how large we take



(a) (b) (c)

Figure 12.1. Some two-loop graphs for Compton scattering. Here straight lines are electrons; wavy lines are photons. The momentum space integral for diagram (a) is convergent, while for (b) and (c) it is divergent, due to the subintegration associated with the subgraphs surrounded by dotted lines.

the various E_f, eventually with enough vertices of type i for which $\Delta_i < 0$, Eq. (12.1.8) will become positive (or zero), and the integral will diverge. Such interactions, whose couplings have negative-definite dimensionality, are called non-renormalizable;[2] theories with any non-renormalizable interactions are also known as non-renormalizable. But this does not mean that such theories are hopeless; we shall see that these divergences may also be absorbed into a redefinition of the parameters of the theory, but here we need an infinite number of couplings.

It should be kept in mind that we have here calculated the degree of divergence of Feynman diagrams arising only from regions of momentum space in which all internal four-momenta go to infinity together. Divergences can also arise from regions in which only the four-momenta of lines belonging to some subgraph go to infinity. For instance, in quantum electrodynamics Eq. (12.1.11) gives $D \leq -1$ for Compton scattering (where $E_e = 2$, $E_\gamma = 2$), and indeed graphs like Figure 12.1(a) are convergent, but a graph like Figure 12.1(b) or 12.1(c) is logarithmically divergent, because these graphs contain subgraphs (indicated by dotted boxes) with $D \geq 0$. We can think of the divergence of these graphs as being due to an anomalously bad asymptotic behavior that occurs when the eight components of the two independent internal four-momenta of these graphs go to infinity on a particular four-dimensional subspace, namely, that subspace in which the only four-momentum actually going to infinity is the one circulating in the loops that are inserted in the internal lines or at an electron-photon vertex.

It has been shown [2] that the requirement for the actual convergence of

[2] In perturbative statistical mechanics, non-renormalizable interactions are called irrelevant, because they become less important in the limit of low energies. Renormalizable and super-renormalizable interactions are called marginal and relevant, respectively.